

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

TypeScript, JavaScript, Python, Java, C++, SQL, Bash, HTML/CSS

Backend & APIs

Node.js, Express, FastAPI, Fastify, REST APIs, async services & queues, integration testing

Security & Performance

Authentication, rate limiting, audit logging, AWS KMS envelope encryption, VAPT remediation, Spark / PySpark tuning, Spark UI, latency profiling

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

Databases & Web

MongoDB, PostgreSQL, Redis, SQLite, React, Stripe

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered a **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines, contributing to **\$340K+** in revenue.
- Owned backend security across the eTriage stack: hardened authentication and rate limiting, implemented **AWS KMS envelope encryption** for sensitive PII / clinical data, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings** to protect customer data.
- Supported production systems at scale: monitored logs during releases and client integrations, diagnosed and shipped fixes for tenant-specific incidents, and added integration tests for regression protection across backend workflows.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, improving type safety and long-term maintainability through targeted refactoring.

Projects

Spark ETL Performance Optimization - NYC TLC (~1.4B rows)

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI, SQL

- Cut end-to-end ETL runtime **~25%** (17–19 min to 14.3 min) on a **~1.4B-row** dataset by replacing shuffle joins with broadcast joins (minutes to ~0.05s), enabling Adaptive Query Execution and skew-join handling, and tuning shuffle partitions, schema-aware reads, and partitioning.
- Reduced analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** via partition and column pruning, filter pushdown, and adaptive skew-join configuration; localized shuffle and skew bottlenecks using **Spark UI** stage-level timings.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Developed a multi-stage agentic web-discovery service that plans typed retrieval, reranks pages, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance for every result.
- Hardened the service with **150+ automated tests**, provider routing and fallback across LLM backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency.

Underdogs Fitness - Production Membership & Payments Platform

underdogsfitness.in

React, Redux, Node.js, Express, MongoDB Atlas, Stripe | solo, live in production 3+ years

- Built and deployed a **production payments and membership platform** solo end-to-end on the MERN stack, with **Stripe + manual-cash hybrid payments and a partial-payment ledger**, membership management, and cron-based renewal automation; moved **90%+** of payments through the portal.
- Designed for multi-branch from day 1 (every entity scoped by `branchId`, JWT-claim auth); when the client opened a 2nd branch a year later, onboarded it as a configuration change with **zero schema migration**; still live in production **3+ years** later.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.